

SIMONA GURRI'

Email ◇ LinkedIn ◇ Homepage ◇ Google Scholar

Postdoctoral researcher specializing in reactive Computational Fluid Dynamics of internal combustion engines fuelled by low-carbon alternatives (H_2 , CNG, ethanol, methanol, ammonia) for heavy-duty applications.

POSITION

Politecnico di Torino, Italy

RESEARCH FELLOW

Mechanical engineering - Reactive Computational Fluid Dynamics - Internal Combustion Engines

Advisor: Alessandro Ferrari

AFFILIATION

2023 - present

Dept. of Energy (DENERG) - Politecnico di Torino

2021 - 2023

Dept. of Environment, Transport and Infrastructure (DIATI) - Politecnico di Torino

INTERESTS

Topics

Internal Combustion Engines · Reactive Combustion Simulation · Low-Carbon Fuels · Powertrain · Transportation systems

Methods

Reactive Computational Fluid Dynamics · Turbulence-Combustion Modelling · Chemical Kinetics & Mechanism Benchmarking · Time Frequency Analysis · Model Based System Engineering

EDUCATION

Ph.D. in Energetics

Italy

Politecnico di Torino

2021 - 2025

- Awarded **cum laude**. Thesis: "Synergies of Bimodal Freight Electric Multiple Units Trains and Advanced ICE Technologies for Sustainable Intermodal Transport" Supervisors: Ferrari A., Dalla Chiara B.
- Italian Cattedra Abertis **Award for best dissertation on sustainable mobility**

M.Sc. in Mechanical Engineering (Erasmus+)

France

EPF Sceaux

2019 - 2020

- Research project: "FEM modeling and simulation in python of the sealing processes of a Nitinol ring for a particle accelerator" Supervisor: Belhadj C.
- Courses held in French.

B.Sc. and M.Sc. in Mechanical Engineering

Italy

Politecnico di Torino

2015 - 2021

- Grade: 110/110 · Thesis: "Functional Testing of a passenger train: the case of the Donizetti regional rolling stock- Systems Engineering approach and basis for Remote Prognostics" Supervisors: Dalla Chiara B., Bonamico M.

PROFESSIONAL EXPERIENCE

RESEARCH FELLOW

B4IA project @ Politecnico di Torino

10/2025 - Present

- Fellowship funded by the *Italian Ministry of Economics*.
- Design and prototyping (TRL 6) of an autonomous bus with a **PFI hydrogen-fueled internal combustion engine**. Focus on engine and injection system (**1D CFD** and **3D CFD** simulations for injectors' and combustion chamber design).
- **1D engine modelling in GT-Power** for H_2 and CNG combustion, including calibration against experimental heavy-duty engine data from *FPT Industrial* and coupling with **3D CONVERGE CFD**.
- **Chemical kinetics analysis in Cantera**: benchmarking of published reaction mechanisms, comparison of computed combustion quantities across mechanisms, and **sensitivity analysis** to support mechanism selection for engine-relevant conditions.
- Collaborating with industrial partners in North-West Italy (*Blue Engineering S.p.A., Italtecnica, De Marco S.p.A., Reply*).

DOCTORAL RESEARCHER

Politecnico di Torino

11/2021 - 10/2025

- Advanced **reactive CFD** modeling of turbulence-combustion interactions in **low-carbon fuel engines** (H₂, CNG, ethanol, methanol). **CONVERGE CFD** employed for 3D U-RANS simulations using **RNG k- ϵ** turbulence model. Validation against engine data provided by *FPT Industrial*.
- Development of a new optimized strategy for **boundary condition setting** in 3D-CFD simulations of engines. Development of a new **spark deposition model for SI engines** in CONVERGE.
- **CINECA HPC Galileo100** cluster and **Polito HPC** efficiently used (considering Amdahl law and resources availability).
- **Time Frequency Analysis** of combustion to set-up a fast predictive real-time model in **Matlab** for engine diagnostics. In partnership with *FPT Industrial*.
- **Transportation-systems research (rail & fleet)**: functional design of a next-generation distributed-power freight train (**System Composer**, **Simulink**, **GaBi LCA**; with *Alstom S.A.*); operation and intermodal-logistics simulation (**OpenTrack**, **Trenissimo**, **Anylogic**, **ARENA DES**; with *RFI* and *SITO*); a **Python** railway microsimulation tool reconstructing lines via **OpenRailwayMap** and **Google Maps APIs**.
- **Data-driven mobility & powertrain analysis**: techno-economic analysis of electric yard tractors (with *PSA Port of Genova*); energy consumption of battery-electric buses across temperatures (*GTT* fleet data); **k-means clustering** of a **200k-vehicle European dataset** (15+ features) to identify driving cycles for targeted powertrain redesign (data from *Stellantis*); EV charging scalability thresholds via **queueing theory**.

TEACHING ASSISTANT

09/2022 - 03/2026

- Taught **140+ hours of frontal classes** on Fluid Machinery, Model Based System Engineering (MBSE) and Transportation Systems in **MSc courses** of Mechanical and Civil Engineering.
- Invited to teach 4 hours on MBSE and data analytics in the **Executive Master on Cableways and Ropeways Systems**.
- Top scores (avg. 3.89/4) on students feedback questionnaires on lecturers performances for 3 years.
- **Supervised 20+ MSc students** for their final thesis in Mechanical, Civil, Energetics or Aeronautical Engineering.

CONTRACT PROFESSOR

Università della Calabria

07/2024

- Delivered course titled "Theory and Numerical Applications through Direct Data Collection from Vehicles" as part of the **postgraduate II level Master's program** in Electric Mobility and Circular Economy (MEES)

RESEARCH SCHOLAR

Blue Engineering S.p.A. & Politecnico di Torino

07/2021 - 11/2021

- Conducted comprehensive analysis on advanced design and maintenance strategies for commuter rolling stock. Employed **Reliability Centered Maintenance (RCM)** principles to develop recommendations for implementing **predictive maintenance techniques** with **data-driven approaches** to improve operational efficiency.

PROCESS ENGINEERING INTERN

Alstom S.A., Savigliano (Italy)

09/2020 - 03/2021

- Implemented **Process FMECA** for Coradia Stream regional train validation, conducting **fault tree analysis** on 15+ critical subsystems. Developed **statistical process control** protocols improving testing efficiency. Collaborated with cross-functional teams to establish **risk priority number (RPN)** matrices for 200+ components, enhancing system reliability metrics.

ADDITIVE MANUFACTURING DESIGNER

Team PACE @PoliTo

09/2017 - 10/2018

- Part of an international team, comprising students from Politecnico di Torino, Tongji and Hunan Universities (China), TU Darmstadt and RWTH Aachen (Germany), and Kookmin University (South Korea). Secured *First Place in Engineering and Industrial Design* at the 19th PACE Forum competition held in Warren, Michigan, USA. More here: PACE.

ACADEMIC SERVICE AND LEADERSHIP

GLOBAL ENGAGEMENT CHAIR

ASME DRIVN 2026 conference

12/2025 - Present

- Conference of the American Society of Mechanical Engineers - Transportation System Division
- Leading international outreach strategy and institutional partnerships to expand global participation and visibility.
- Conducting a quantitative survey analysis to improve engagement over the years

VICE CHAIR

ASME Italy Section

08/2026 - Present

- Contributing to the development of ASME activities in Italy, supporting professional engagement, networking, and initiatives for the Italian engineering community.

PEER REVIEW ACTIVITIES

Journals 2026 - present

- **Energies**, MDPI

PEER REVIEW ACTIVITIES

International Conferences 2022 - present

- **IEEE ITS Conference Naples 2026** (Intelligent Transport Systems & Logistics)
- **AIIT TIS Conference Rome 2024** (Next-Gen Mobility Systems)
- **SAE Conference Torino 2024** (CO₂ Reduction Technologies)
- **AIIT TIS Conference Rome 2022** (Transport Infrastructure & Systems)

PUBLICATIONS AND PROCEEDINGS

Journal Publications

- 2026 **Gurri S.**, Vento O.*, Marello O., Finesso R., Gallo A., Vargiu L., Ferrari A., *A Methodology for boundary condition optimization in 3D-CFD of internal combustion engines*, *Applied Thermal Engineering* - PUBLICATION HERE: DOI
- 2026 Olivari E., **Gurri S.***, De Marinis F., Caballini C., *Simulation-based decision-making framework for sustainable intermodal transport including next-generation freight electric multiple-units*, *Case Studies on Transport Policy* - PUBLICATION HERE: DOI
- 2025 De Marinis F., **Gurri S.***, Dabove P., *Development of a GNSS positioning system to support automatic manoeuvring of new-generation freight trains at low speeds*, *Ingegneria Ferroviaria* - PUBLICATION HERE: DOI
- 2025 **Gurri S.**, Cappelli F.*, Dalla Chiara B., *Matching demand and supply in motorised mobility: a data-driven differentiation of the driving patterns for urban contexts*, *Transportation Engineering* - 2025, PUBLICATION HERE: DOI
- 2024 Ferrari A., **Gurri S.**, Vento O.*, *Injected Fuel Mass and Flow Rate Control in Internal Combustion Engines: A Systematic Literature Review*, *Energies* - 2024, PUBLICATION HERE: DOI
- 2024 Olivari E., **Gurri S.**, Caballini C.*, Carotta T., Dalla Chiara B., *Ports go green: an energy-based cost analysis for evaluating the electrification of yard tractors*, *The Open Transportation Journal* - 2024, PUBLICATION HERE: DOI
- 2023 **Gurri S.***, Bocchieri M., Galasso D., Operti V., Dalla Chiara B., *Assessing the speed of an electric multiple-unit freight train on high-speed lines*, *Ingegneria Ferroviaria* - 2023, PUBLICATION HERE: DOI

Conference Contributions

- 2026 Olivari O., Gurri S., De Marinis F., Caballini C., *Energy-efficient joint optimization of train scheduling and track assignment in intermodal freight terminals*, IEEE Intelligent Transportation Systems Conference (ITSC), September 15-18, 2026 (Naples, Italy) - DOI TO BE DEFINED
- 2025 Olivari E., Gurri S., De Marinis F., Caballini C., *A Hybrid Simulation Framework Combining DES and Microsimulation for Multicriteria Evaluation of Intermodal Freight Systems: A Case Study on Next-Generation Trains*, TransClim 2025 (Transport Systems and Climate Change), June 18, 2025 (Livorno, Italy) - NO PROCEEDINGS
- 2024 Gurri S., Zara G., Di Paola A., Dalla Chiara B., *Design of an integrated monitoring system for multiple-unit freight trains*, 6th International Conference on Railway Technology, September 1-5, 2024 (Prague, Czech Republic) - PUBLICATION HERE: DOI
- 2024 Gurri S., Cappelli F., Dalla Chiara B., *Matching demand and supply in motorised mobility via data analysis*, SIDT Conference, June 12-15, 2024 (Villasimius, Italy) - NO PROCEEDINGS
- 2024 Gurri S., Olivari E., Caballini C., Dalla Chiara B., *Next-generation freight intermodal trains: an effective way to increase rail share and sustainability of inland transportation*, SIDT Conference, June 12-15, 2024 (Villasimius, Italy) - NO PROCEEDINGS
- 2023 Gurri S., Zara G., Di Paola A., Dalla Chiara B., *Preliminary design of an integrated monitoring system for multiple-unit freight trains*, UIC High Speed Rail, March 07-10, 2023 (Marrakesh, Morocco) - NO PROCEEDINGS
- 2022 Gurri S., Santacaterina E., Guarrera M., Dalla Chiara B., *Driving modal shift on low-traffic railway lines through technological innovation*, Transportation Research Procedia - AIIT International Conference TIS, September 15-16, 2022 (Rome, Italy) - PUBLICATION HERE: DOI
- 2022 Gurri S.; Bocchieri M.; Galasso D.; Operti V.; Dalla Chiara B., *Simulating the operation of new-generation freight-EMUs for high-speed lines: perspectives for more reliable, sustainable, and fast logistics*, 5th International Conference on Railway Technology, August 22-25, 2022 (Montpellier, France) - PUBLICATION HERE: DOI

Under Review

- 2026 Gurri S., Vento O., Ferrari A., *A Physics-Based Spark Discharge Model for Simulation of Cycle-to-Cycle Variability in Internal Combustion Engines*
- 2026 Gurri S., Ferrari A., Vento O., *0D-to-3D Benchmarking of Chemical Mechanism Selection on Combustion Simulations of Heavy-Duty Engines*
- 2025 Sardella G., Gurri S., Andrenacci N., Dalla Chiara B., Vellucci F., Pede G., Zanini M., *BEB for Public transport: a methodological approach and data resulting on energy consumption at different temperatures*

Dissemination

- 2023 Dalla Chiara, B.; Gurri, S., *Efficienza energetica nel trasporto merci: quali strumenti.*, Logistica Management, pp 54-61 - MARCH 2023

HARD SKILLS

CFD & Engine Sim	CONVERGE CFD, GT-Power
Chemical Kinetics	Cantera (mechanism benchmarking, sensitivity analysis)
HPC	HPC systems (MPI, OpenMP)
Coding & Data	Python (NumPy, SciPy, pandas, ...), Matlab, Excel
CAD	Solidworks, Inventor, Autocad 2D, NX11
Rail Simulation	Arena, Anylogic, OpenTrack, Trenissimo

COURSES, CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- 2026 **AI/ML for Fluid Dynamics**
22 hours, *Flow Thermo Lab*
- 2026 **Mentee, Young ISSNAF Mentoring program**
Selected participant in international mentorship program connecting Italian researchers with senior scientists in North America through *ISSNAF*
- 2024 **Dive into the World of HPC: Parallel and Accelerated Computing**
80 hours, *CINECA*
- 2023 **Towards Data Analysis and Machine Learning – Data Structures in Python (Advanced Level)**
40 hours, *Politecnico di Torino*
- 2021 **Railway Engineer Course – Organization, Technics, and Safety in the Railway Field**
60 hours, *CIFI*
- 2021 **Graduate Record Examination (GRE) – General Test**
ETS
- 2019 **TFI – Test de Français International**
ETS EMEA
- 2018 **CSWA – Certified SOLIDWORKS Associate in Mechanical Design**
SolidWorks Authorized Training Centre

AWARDS

- 2026 ^{4th} Italian Cattedra Abertis **Award for PhD research** on sustainable mobility (5000 €) – [More here](#)
- 2018 **First place in Engineering and Industrial Design** at 19th PACE Forum, Warren, Michigan, USA – [More here](#)

ADDITIONAL INFORMATION

Languages	Italian (native) · English · French · Spanish (basic)
Licenses	Professional Engineer (Italy) · Driver's License · Scuba Diving Certification (Open Water)
Interests	Marathon training · Hiking · Mountain biking · Photography · Padel, tennis & volley · Literature